Protocol for Generating Viscosity Maps in Planktonic Systems Using Multiple Particle Tracking Microrheology (MPTM)

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This document outlines the steps needed to measure and map the zero-shear dynamic viscosity in planktonic systems with a spatial resolution down to 2µm, using Multiple Particle Tracking passive Microrheology (MPTM). While this methodology can be extended to explore non-Newtonian rheological behaviors, we assume Newtonian fluid properties throughout this protocol.

This protocol was developed in the Physical Ecology Lab at the University of Lincoln and was used to generate the maps published in Guadayol, Segura-Noguera, and Humphries (2020) and Guadayol et al. (2021).

Introduction

Viscosity in the microbial world

Viscosity is defined as a fluid's resistance to deformation, and it plays a dominant role in the lives of microscopic organisms. At the microscale, the inertial forces that shape our everyday experience of our environment, are negligible. Instead, viscous forces dominate. To illustrate this, picture a 1µm long bacterium swimming in water at 30µm/s. Its inertia is so low that were it to stop spinning its flagella, it would take only 0.3 µs for the bacterium to come to a halt, and would coast for only 0.1Å (Purcell 1977). That is 1/10 of the diameter of a hydrogen atom!

The relative importance of inertia and viscosity is classically captured by the non-dimensional Reynolds number, defined as:

$$\mathrm{Re} = \frac{\rho v L}{\eta}$$

where ρ is the density of the fluid, v is the velocity of the flow, η is the dynamic viscosity and L is the representative length scale of the flow, which could be, for example, the width of the channel or the diameter of the swimming microorganism, depending on your question. In essence, *Re* is the ratio of inertial and viscous forces. Microorganisms usually experience very low *Re* regimes. This means that:

1. Inertia is negligible to microbes.
2. Microscopic particles and molecules experience Brownian motion, that is, the jiggering motion that is a consequence of innumerable collisions with water molecules. For motile organisms, this disrupts straight-line movement and resource tracking; for non-motile particles, it drives diffusion.
3. Transport of molecules (e.g., nutrients or info-chemicals) is ultimately realized by molecular diffusion, not by advection.

4. There are constraints on how microorganisms can accomplish active movement, as simple, reciprocal movements such as those of a swimming scallop are ineffectual at the microscale. In other words, unlike in inertial systems, there is **kinematic reversibility**. This is why, for example, bacteria use corkscrew flagella or eukaryotic flagellates use asymmetric appendage motions.

What is microrheology

Rheology is the study of how materials deform under applied force. Thus, it deals with concepts such as viscosity and elasticity. In fluids, these properties are usually measured in bulk, using instruments like rheometers or viscometers which require relatively large sample volumes, typically in the milliliter range. This is clearly not very practical when we are interested in the environment experienced by single microorganisms.

Microrheology is a set of techniques, complementary to classical rheology, that allow us to measure rheological properties of materials at much smaller scales. There are quite a few different microrheological techniques, including, for example, multiple particle tracking, dynamic light scattering, diffusive wave spectroscopy, differential dynamic microscopy, optical trapping or magnetic bead microrheology. Despite their differences, all these techniques seek to characterize the motions of colloidal spherical particles suspended in a fluid in order to infer its rheological behaviour. For a comprehensive review of microrheological techniques and their theoretical base and practical uses, we recommend Furst and Squires (2017).

Microrheological techniques can be broadly categorized into two types: those that analyze the thermal motion of particles (i.e., Brownian motion), known as "passive microrheology," and those that impart a known force to particles and measure their response, referred to as "active microrheology." The specific technique we describe in this protocol is a passive method called Multiple Particle Tracking Microrheology (MPTM).

Principles of passive microrheology

In passive microrheological techniques, we take advantage of the Brownian motion of particles suspended in a fluid this random motion arises from countless collisions with solvent moleculesand was first observed and described by botanist Robert Brown in the first half of the XIX century. The Brownian motion of a suspended spherical particle can be described by the Stokes-Einstein equation, that relates the diffusive motion of the particle with its resistance to motion and the thermal energy of the fluid:

$D = \mathrm{k_B}T/\zeta$

,where D is the diffusion coefficient (units $\mathrm{L^2T^{-1}}$), the thermal energy is parameterized by the product of the Boltzmann constant $\mathrm{k_B}$ and the fluid temperature T (K), and ζ is the particle's resistance to motion.

If you recall, we defined viscosity as the resistance of a fluid to be deformed. Thus, ζ must be related to viscosity. This relation is made explicit in the Stokes's law, which describes the resistance to motion of a spherical particle:
$\zeta = 6\pi\eta r$
where η is the dynamic viscosity of the fluid, and r is the radius of the spherical particle. Combining both equations we obtain the Stokes-Einstein equation, which describes the diffusivity of a spherical

particle suspended in a Newtonian fluid as a function of the temperature and viscosity of the fluid and and the radius of the particle.

$D = k_B T / 6\pi\eta r$

Thus, in a nutshell, passive microrheology involves tracking spherical, approximately neutrally buoyant particles dispersed on a fluid to determine their diffusivity. Given known temperature and radius, the viscosity of the surrounding fluid can then be estimated from Stokes-Einstein.

Colloidal probes

The spheres used in passive microrheology need to fulfill certain conditions to ensure accurate measurements. They have to be small enough to ensure Brownian motion is easily characterized, but still much larger than the solvent molecules so that the fluid can be considered a continuum, and to be easy to track. They have to be spherical, very uniform in size and shape, and to show a high contrast, which is why fluorescent microspheres are most commonly used. Their density needs to be as close as possible to the solvent fluid to minimize sedimentation, and they need to be chemically stable and hydrophilic to minimize aggregation driven by van der Waals and electrostatic forces. The colloids that best meet these requirements are fluorescent carboxyilate polystyrene microspheres, although other materials such as inorganic silica particles may also be suitable depending on the application.

Reagents and material

- Polystyrene fluorescent microspheres. In order to minimize aggregation, particularly in high ionic strength seawater, we recommend using PEG-modified beads, which can be manufactured on special order. We obtained ours from microParticles GmbH (Berlin, Germany). Beads can be as large as 1µm in diameter and as small as your system can efficiently track, and the choice of fluorocrome should depend on your particular setup. We use Yellow-Green beads with a diameter of 0.29µm.

- Surfactant (Tween 20) (only if non PEG-modified beads are used):
Tween20 (P2287 TWEEN 20, Sigma Aldrich)
https://www.sigmaaldrich.com/ES/es/product/sial/p1379

- Sodium azide
https://www.sigmaaldrich.com/ES/en/product/sial/s2002
Not critical. This is to arrest movement of bacteria in the samples and/or to stop unwanted bacterial growth in the beads solution for long-term storage.

- A commercial polymer for the calibration. We have always used Ficoll PM70 (https://www.sigmaaldrich.com/ES/es/product/sigma/ge17031010) or Ficoll PM400, but dextran, glycerol or polyethylene glycol (PEG) can also be used. It is critical that the solution is approximately Newtonian and that the functional relationship between V/V and viscosity is well described and quantified in the literature.

- Filtered seawater, artificial seawater or freshwater, depending on your sample.

- Poly-lysine-coated slides. For example:
https://uk.vwr.com/store/catalog/product.jsp?catalog_number=631-1349
In case long-distance objectives are not available, coverslips need to be coated instead.

- Secure-Seal™ Adhesive Spacers
  https://www.sigmaaldrich.com/ES/en/product/sigma/gbl654004
  This ensures no drift and uniforme height of the sample
- Coverslips.
- Automatic pipettes with tips.
- A few 2mL cryovials

Equipment

- An epifluorescence microscope, ideally inverted equipped with a 40X long distance objective.
- An optical table, to minimize vibrational noise. If this is not available, additional calibration will be needed to determine the error.
- A camera capable of recording at 50 FPS or more.
- Temperature-controlled stage.
- A digital thermometer to measure the temperature at the stage. The best would be to use a temperature controller microscope stage, if you have it. But the essential thing is not to control temperature, but to know what is the temperature at the time of sampling.
- Microcentrifuge.
- Vortex.
- Bath sonicator.
- Computer with MATLAB
- To prepare calibration standards
  - A shaker or a stirrer
  - Scale

Steps

Preparation of a working suspension of beads

1. **Preparation of surfactant solution**
   1. Dilute Tween 20 into 500 mL of artificial seawater or of aged seawater sterilized by filtration to a final concentration of 0.05% W/V. Tween 20 prevents beads from aggregating.
      Tip: Tween 20 is very viscous and it is difficult to accurately pipette a precise volume. It is best to measure it by weight than by volume.
   2. Add sodium azide (NaN_3) to a final concentration of 80 µM. This will prevent bacterial growth and arrest microbial motility, which could later interfere with microrheology.

2. Add 1 drop (~40µL) of the stock suspension of beads to 100µL of the surfactant solution in a cryovial. If the volume fraction ϕ of the stock suspension is 2.7% w/V, the working suspension will be 2.7*40/140 = 0.77%.

3. Briefly vortex to mix and sonicate for 5 minutes in a bath sonicator at room temperature.

   Tip: It is recommended to wash away the surfactant after this step (Furst and Squires 2017). This is done by successive centrifugations. The objective is to change the surface chemistry of the microspheres to avoid them to aggregate, while minimizing the amount of surfactant in the solution, which could alter the rheological measurements. However, in our experience with seawater samples, washing away the Tween20 results in very high aggregation of beads. This is caused by the high ionic strength of salty water. Therefore, our strategy has been to minimize the volume of beads solution added to the sample.

4. Cover cryovial in tin foil and label it.

5. Prepare a plain slide with a diluted sample of the freshly made suspension to check that beads are not aggregated. Repeat previous step if they are not. If after repeating beads are still aggregated, consider increasing the concentration of surfactant in the solution or using a brand new stock of microspheres.

   Tips: The surfactant solution can be stored at 4C in the darkness for a few weeks. The beads suspension can be reused for a few days as long as it is stored at 4 C in the dark and it is sonicated and checked for aggregates before every session.

Sample preparation

1. **Sample seeding**. Add a volume of the beads suspension to an aliquot of your sample. Final concentration of microspheres should be chosen in relation to the average horizontal distance between contiguous microspheres *d*. This can be estimated using:
   $d \sim C_{2D}^{-1/2}$
   , where C_{2D} is the concentration of beads in the optically sampled plane (beads/µm²), and is the product of the 3D concentration of beads C_{3D} and the depth of field DOF. C_{3D} is calculated as:

   $$C_{3D} = \frac{\phi * df}{4/3 \pi a^3 \rho}$$

   where ϕ is the volume fraction, df is the dilution factor, a is the radius of the microspheres and ρ is their density. Thus the df needed for a given d can be calculated as:

   $$df = \frac{4/3 \pi a^3 \rho}{\phi DOF d^2}$$

   For example, let's calculate the *df* needed to obtain a suspension of microspheres with and average distance *d* of 4 µm. Assuming a *DOF* of 2.6µm, reasonable for a 40X air objective with blue light, and a ϕ=0.77% for the working beads solution, and using polystyrene microspheres of 0.29 µm in diameter and ρ=1.05 g/mL, *df* should be ~0.074, The volume of beads working solution to add to 100µL of sample would be: V=0.074/(1-0.074)X100 = ~4µL

2.

   of adding 4µL of a suspension of 0.29µm beads with a volume fraction of ϕ= 1% into 100µL sample would result in a final volume fraction of ϕ= 0.01*4/104 = 0.04%. Given a V_p = $1/6 \cdot \pi \cdot 0.29^3 = 1.28 \cdot 10^{-2}$ µm³ would result in an average distance *d* of ~3µm.

   Critical tip: Do not mix! That could destroy the very structures you want to study. Wait a few minutes for the beads to diffuse across the entire volume.

3. **Sample mounting**. Very gently pipette the sample onto a poly-lysine-coated slide. To minimize wall effects, sample can be placed in between two coverslips that will raise the height of the chamber. Top it with a coverslip taking care not to introduce air bubbles. Seal with vaseline or nail polish to suppress any drift. The sample is now ready for visualization.

Sample visualization

1. Set up the inverted microscope with the appropriate objective lens for imaging the microspheres. Connect camera to the microscope and ensure it is properly calibrated for the particular objective you will use.
   For example, 0.29 microspheres can be readily visualized in an inverted epifluorescence microscope with a x40 LD distance objective.

2. Set up temperature stage to a consistent working temperature. If stage is only heating, set a temperature somewhat higher than room temperature.

3. Place the sample on the microscope stage.

4. Adjust the illumination settings to optimize the visibility of the microspheres.
   Use the minimum light intensity that allows clear visualization of the fluorescent microspheres. High light intensities for long periods of time may harm microorganisms or affect their behaviour. Adjust camera gain if necessary.

5. Focus on an object of interest (e.g. a phytoplankton cell, an exopolymeric particle or an aggregate) that is attached to the slide. Focus ~10µm away from the slide surface to minimize bias caused by hydrodynamic interactions of the microsphere with the slide surface. Observe for a few minutes to ensure the object of interest is not moving.

Data acquisition

1. Capture one or several time-lapse videos of the particle motion over a predetermined duration (e.g., 10-30 seconds) at a suitable frame rate (e.g., 30-100 frames per second). Take several videos at a minimum frame rate of 30 fps.

2. **Critical step**: Before and after each video, record a still image under brightfield illumination that clearly shows the particle of interest. This is to 1) document the object of interest and 2) ensure there is no motion or drift that could bias results.

Tip: optimal frame rate depends on the size the microspheres and the pixel size (and hence on the resolution of the camera and the magnification of the objective). In our experimental setup, with microspheres of 0.29µm in diameter and a pixel scaling of 0.378, we obtained robust results with a frame rate of $30s^{-1}$. To study rheological responses at very high frequencies, it may be desirable to work at higher frame rates.

Tip: To reduce video file size, consider saving with 8-bit color depth and reducing the frame dimensions.

Data analysis

Microsphere tracking. The microsphere tracking used a Matlab code initially designed to track bacterial cells (Guadayol 2016).

Viscosity estimation. All the code, including the code used to estimate local relative viscosities and create the maps, is available online under a GPL 3.0 license (Guadayol 2020). Briefly, the field of view is segmented into squares of user-defined side. For each individual microsphere trajectory, squared displacements (SD) with lag-times τ ranging between $1/f$ (where f is the frame rate) and a user-defined maximum value are computed. SD are grouped into the square partitions according to their mean centroid, and the ensemble Mean Square Displacements (MSDs) are computed for each square. The diffusivity D of the microspheres within each square is calculated by least squares fitting the data to the equation $\mathrm{MSD}(\tau) = 2dD\tau$, where $d = 2$ is the dimensionality of the tracks. The average dynamic viscosity η for each position is then calculated from the Stokes-Einstein equation for diffusion of a spherical particle $D = k_\mathrm{B}T/(6\pi\eta a)$, where k_B is the Boltzmann constant and T is the absolute temperature. Relative viscosities are calculated dividing local dynamic viscosities by temperature and salinity dependent viscosity of ASW estimated using equations in (Sharqawy et al. 2010).

Calibration and QC

Before performing any microrheological measurement, it is advisable to carry out an initial thorough calibration of the setup. This is needed in principle only once, unless significant changes are made to the original setup. Its purpose is to detect and correct potential sources of error, such as unwanted vibrations, temperature fluctuations, or variations in light intensity. The calibration involves performing microrheology on a series of solutions of Newtonian fluids of known viscosity. We present here the protocol for using Ficoll, a synthetic, high-molecular-weight polysaccharide derived from the fermentation of sucrose by the bacterium *Methylobacterium extorquens* that is often used to create density gradients for the separation of cells, organelles, and other biological particles based on their size and density. However, other Newtonian polymers may also be used for calibration.

Preparation of Ficoll 70PM calibration standards

1. **Determine the desired concentrations of polymer.** These should be informed by the viscosity vs. concentration curve published by the polymer maker and the expected range of viscosities in your experiments. We recommend 5 solutions. A final volume of 5mL is enough to conduct the microrheology measurements.

2. **Calculate the amount of polymer needed**:

   Use the formula:
   Amount of polymer (g) = Desired concentration (g/mL)×Final Volume (mL)

   - For example, for a 10% w/v solution of Ficoll in 5 mL DI water:
     Amount of Ficoll = (10g/100mL×5mL)=0.5g

3. **Weigh the Ficoll**: Use a balance to accurately weigh the calculated amount of polymer and put it onto the containers.
   TIP: Ficoll comes in a fine white powder that gets very sticky once wet. Try adding all the powder to the flasks or cylinders before putting the water. Washing out the weighting boats is going to be tricky.

4. **Add Water**:

    - Place the polymer in volumetric flasks or graduated cylinders.
    - Add a portion of the same water used in your experiments (about 50-70% of the final volume) to the Ficoll. This helps dissolve the Ficoll more easily.
5. **Mix the Solution**:
    - Stir the mixtures using a magnetic stirrer or placing them in a shaker overnight or until the Ficoll is completely dissolved. Ficoll can take some time to dissolve, so ensure that there are no undissolved particles. Seal the containers properly to avoid evaporation.
6. **Adjust the Volume**:
    - Once the Ficoll is fully dissolved, add more water to bring the total volume up to the desired final volume (e.g., 5 mL for a 10% solution).
7. **Final Mixing**: Stir the solution again to ensure it is homogeneous.
8. **Storage**: If not used immediately, store the Ficoll solution in a properly labeled container at the appropriate temperature (usually at 4°C for short-term storage).
9. If available, **measure bulk viscosity** of the prepared solutions using a rheometer or viscometer.

Data Consistency Check

We recommend taking at least one measurement of milliQ water every day to ensure consistency across days.

Troubleshooting

Poor spatial coverage in the viscosity map. The coverage of the map depends heavily on the concentration of beads and the duration of the video recording. Higher concentrations and longer videos improve the coverage. However, these parameters involve trade-offs:

Bead concentration: It is important to use the minimum volume of beads suspension to avoid significantly diluting the sample, which could alter its rheology. Based on our experience, a final concentration of 0.29µm beads ø10^9 beads·mL^{-1} , corresponding to a volume fraction of ø10^5, yields high quality maps.

Video duration: Longer recordings consume more memory and may become impractical for data processing and analysis. I

Microspheres aggregation. The high ionic strength of seawater promotes aggregation of colloids. The usage of PEG-modified microspheres greatly decreases this problem. However, if you are using regular plain or carboxylated PS microspheres, you can try the following steps:

1. Add an anionic surfactant, such as Tween 20 to the working solution of beads. Typical final concentrations of Tween 20 are 0.05% w/V.
   Tip: Tween 20 is very viscous and it is difficult to accurately pipette a precise volume. It is best to measure it by weight than by volume. It is easiest to prepare a large volume of surfactant solution as a base for the beads working solution. For example, dilute Tween 20 into 500 mL of artificial seawater or of aged seawater sterilized by filtration to a final concentration of 0.05% w/V and add add sodium azide to a final concentration of 80 µM to allow long-term storage in the fridge.

2. Decrease the density of the microspheres suspensions.
3. Use a fresh batch of beads.

Microspheres sedimentation. Polystyrene density (1.05 $g \cdot mL^{-1}$) is slightly higher than that of seawater (1.024 $g \cdot mL^{-1}$ for 33.5 salinity at 20ºC). Under these conditions, microspheres would sink on average at ~1$nm \cdot s^{-1}$. In freshwater faster sedimentation can become an issue. To mitigate this effect, yo may consider lowering the temperature of the fluid or using smaller microspheres.

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